

DAT535 – Data Engineering Fall 2026



University
of Stavanger

Medallion Architecture Data Engineering Lifecycle

Week 36 reminder

- <https://wayground.com/join?gc=17687013>

Table 3: Top 40 occupations with highest AI applicability score.

Job Title (Abbrev.)	Coverage	Cmpltn.	Scope	Score	Employment
Interpreters and Translators	0.98	0.88	0.57	0.49	51,560
Historians	0.91	0.85	0.56	0.48	3,040
Passenger Attendants	0.80	0.88	0.62	0.47	20,190
Sales Representatives of Services	0.84	0.90	0.57	0.46	1,142,020
Writers and Authors	0.85	0.84	0.60	0.45	49,450
Customer Service Representatives	0.72	0.90	0.59	0.44	2,858,710
CNC Tool Programmers	0.90	0.87	0.53	0.44	28,030
Telephone Operators	0.80	0.86	0.57	0.42	4,600
Ticket Agents and Travel Clerks	0.71	0.90	0.56	0.41	119,270
Broadcast Announcers and Radio DJs	0.74	0.84	0.60	0.41	25,070
Brokerage Clerks	0.74	0.89	0.57	0.41	48,060
Farm and Home Management Educators	0.77	0.91	0.55	0.41	8,110
Telemarketers	0.66	0.89	0.60	0.40	81,580
Concierges	0.70	0.88	0.56	0.40	41,020
Political Scientists	0.77	0.87	0.53	0.39	5,580
News Analysts, Reporters, Journalists	0.81	0.81	0.56	0.39	45,020
Mathematicians	0.91	0.74	0.54	0.39	2,220
Technical Writers	0.83	0.82	0.54	0.38	47,970
Proofreaders and Copy Markers	0.91	0.86	0.49	0.38	5,490
Hosts and Hostesses	0.60	0.90	0.57	0.37	425,020
Editors	0.78	0.82	0.54	0.37	95,700
Business Teachers, Postsecondary	0.70	0.90	0.52	0.37	82,980
Public Relations Specialists	0.63	0.90	0.60	0.36	275,550
Demonstrators and Product Promoters	0.64	0.88	0.53	0.36	50,790
Advertising Sales Agents	0.66	0.90	0.53	0.36	108,100
New Accounts Clerks	0.72	0.87	0.51	0.36	41,180
Statistical Assistants	0.85	0.84	0.49	0.36	7,200
Counter and Rental Clerks	0.62	0.90	0.52	0.36	390,300
Data Scientists	0.77	0.86	0.51	0.36	192,710
Personal Financial Advisors	0.69	0.88	0.52	0.35	272,190
Archivists	0.66	0.88	0.49	0.35	7,150
Economics Teachers, Postsecondary	0.68	0.90	0.51	0.35	12,210
Web Developers	0.73	0.86	0.51	0.35	85,350
Management Analysts	0.68	0.90	0.54	0.35	838,140
Geographers	0.77	0.83	0.48	0.35	1,460
Models	0.64	0.89	0.53	0.35	3,090
Market Research Analysts	0.71	0.90	0.52	0.35	846,370
Public Safety Telecommunicators	0.66	0.88	0.53	0.35	97,820
Switchboard Operators	0.68	0.86	0.52	0.35	43,830
Library Science Teachers, Postsecondary	0.65	0.90	0.51	0.34	4,220

Note: Metrics reported as mean of user goal and AI action score.

What data scientists spend most of the time doing?

● Building training sets
 ● Cleaning and organizing data
 ● Collecting data sets
 ● Mining data for patterns
 ● Refining algorithms
 ● Others

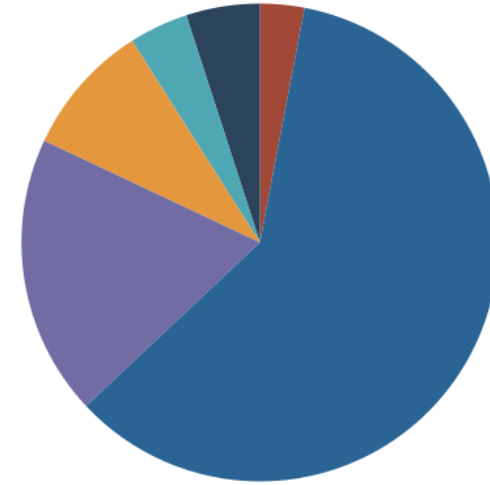


Image taken from forbes.com

Medallion Architecture

- Lifecycle is tool-agnostic; Medallion is usually implemented with Spark + Delta Lake or MSFT Fabric
- Both aim to ensure data quality, scalability, and governance, but Medallion adds a physical structure to enforce these principles.



Lifecycle Stage	Medallion Layer	Purpose
Generation & Storage	Bronze	Raw, ingested data (as-is)
Ingestion & Early Transform	Silver	Cleaned, standardized, conformed data
Transformation & Serving	Gold	Aggregated, business-ready datasets

Delta Lake

Delta Lake is an **open-source storage layer** that brings **ACID transactions, schema enforcement, and time travel** that supports Medallion Architecture.

Key Features

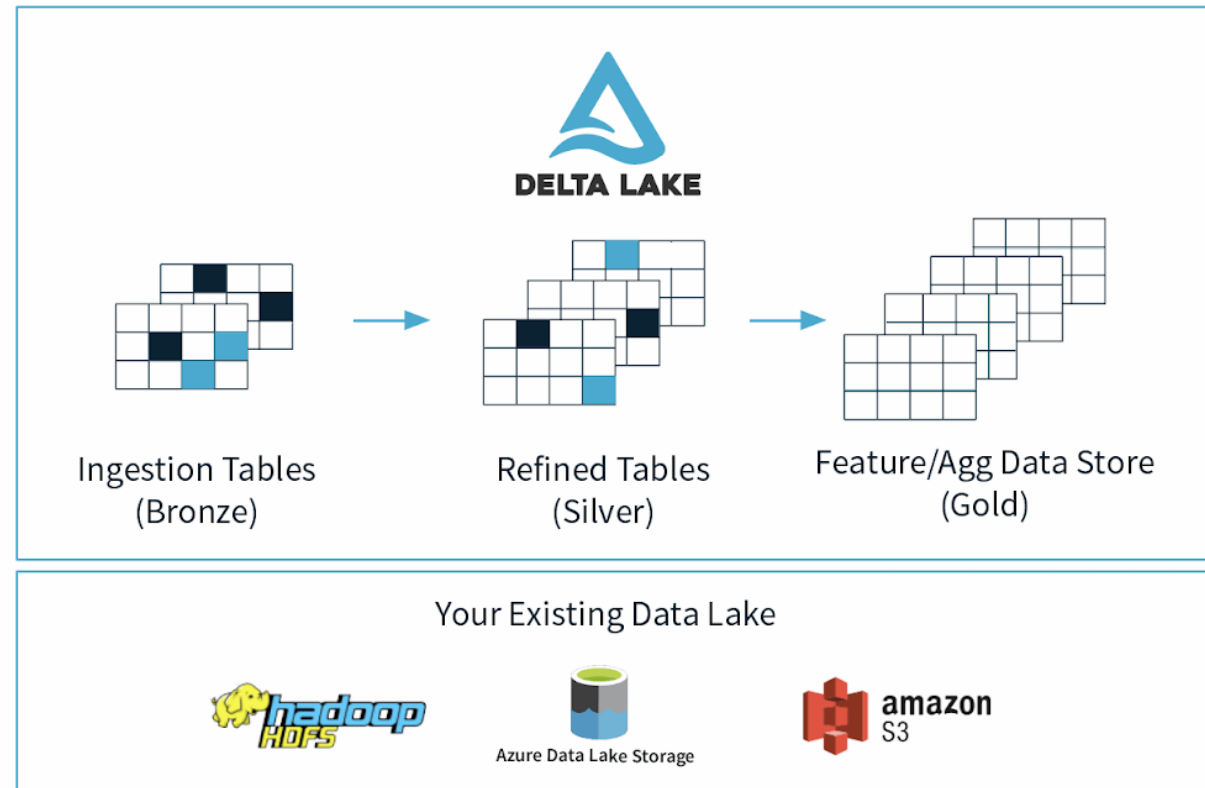
- **ACID Transactions:** Reliable writes and concurrent reads.
- **Schema Enforcement & Evolution:** Prevents "data swamp" issues.
- **Time Travel:** Query historical versions of data.
- **Unifies Batch & Streaming:** Same table for both ingestion modes.



Streaming



Batch



Analytics
and Machine
Learning

Data Engineering Lifecycle and Role

Provides a **holistic framework** for data engineering beyond tools.

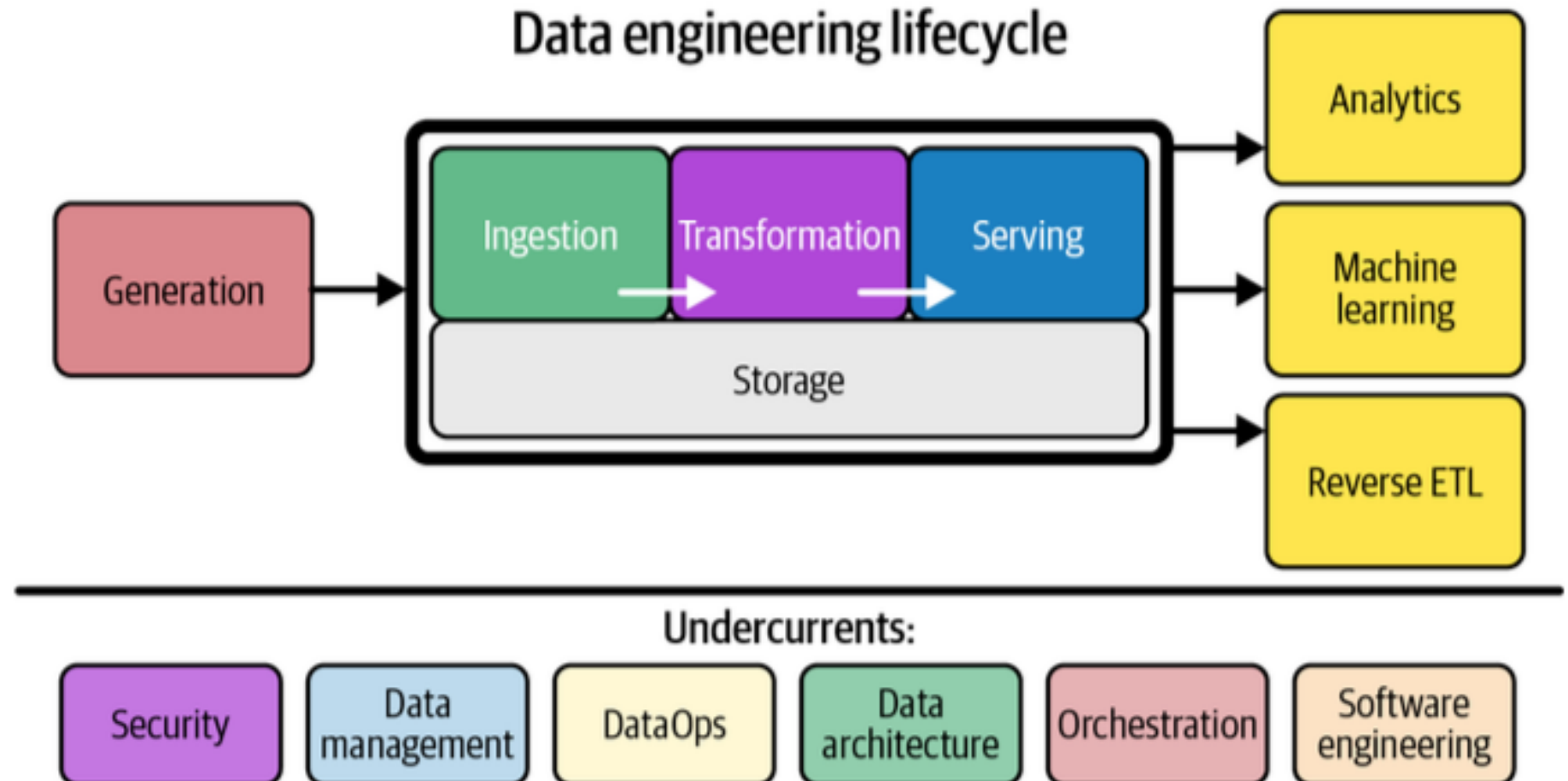
Focuses on **data flow and value creation**, not just technology stacks.

Helps engineers **think in terms of principles**, not just tools.

Ensures **end-to-end accountability** for data quality and reliability.

Bridges the gap between **data engineering and business value**

Storage plays a central role – **discuss why?**



Discussion

1. Why not put everything directly into Gold?
2. Why keep raw data forever?
3. Could a small company skip Silver?
4. Where does AI fit?
5. What is the difference between storage and serving?
6. Which layer fails most often?

Summary

Data Engineering Lifecycle

- Concerned with the **flow of work and responsibilities**.
- Covers the entire journey from data creation to consumption.
- Includes people, governance, architecture, operations, and technology choices.

Medallion Architecture

- Concerned with the **organization of datasets** inside a platform.
- Primarily a storage and transformation pattern.
- Focuses on Bronze, Silver, Gold data layers.